Mahadev Sunil Kumar

PhD Studentship in Computer Science - Algorithms/Hardware Co-Design for Bayesian and Ising Models

Reference: R143025

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October 21, 2026

Dear Prof. Rajendran and Members of the Selection Committee,

I am writing this to apply for the *PhD studentship in Algorithms/Hardware Co-Design for Bayesian and Ising Models* (Reference: R143025) at Northeastern University London in collaboration with IMEC. I hold a Bachelor of Technology in Computer Science with a specialization in Artificial Intelligence from Amrita Vishwa Vidyapeetham (Amrita University), and I currently work as a Research Assistant at the SustainAI Lab at the Indian Institute of Technology, Guwahati. My research has focused on efficient machine learning, neuromorphic computing, and the relationship between computational models and the hardware on which they run. I am interested in this studentship as it would allow me to research and learn these problems in the context of probabilistic computing, where the physical properties of emerging devices can directly influence how algorithms are designed and run.

My research on oscillator-based neuromorphic computing has given me experience learning how the inherent dynamics of physical systems can be used to perform computation. In my work, *Neuromorphic Computing Based on Parametrically Driven Oscillators and Frequency Combs*, which is in press at *Applied Physics Letters*, I developed a physical reservoir computing system based on parametrically driven nonlinear oscillators and frequency combs. I constructed a 2048 dimensional reservoir and evaluated its performance on standard chaotic

Sincerely,
Mahadev Sunil Kumar
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